The Barn Effect: Domain and Multi-Modal Extensions

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Abstract

This companion note extends the core Barn Effect algorithm (see `barn_effect.tex`) to non-tabular domains and multi-modal embedding settings. The two-phase structure—discretize, accumulate, attribute—generalizes to any domain by substituting the encoding step; the accumulator, inference, and convergence results carry over unchanged. Sections 1–4 cover text, image, retrieval-augmented generation, and multi-modal settings. Section 5 treats the multi-modal embedding case in detail, including clustering-based bin schemas and cross-modal gap analysis.

1 Domain Extensions

The Barn Effect is stated for tabular data, where features are independent scalar-valued columns and rows are independent samples. The same two-phase structure generalizes to other domains by changing only the encoding step; the accumulator update rules, inference, and the convergence result carry over unchanged.

1.1 Text / NLP

Each document $d^{(n)}$ is a sample; its tokens $w_1, \ldots, w_T$ are the features. The pairwise structure is *sequential rather than all-pairs*: each token w_k is paired only with its immediate predecessor w_{k-1} and successor w_{k+1}, yielding three interaction lookups per token position (standalone, next-bigram, prev-bigram). The outcome is parametric: $Y_{w_k}^{(n)} := (\hat{f}(d^{(n)}) - \bar{f}) \cdot \mathrm{tfidf}(w_k, d^{(n)})$, where $\hat{f}$ is any scalar model and the TF-IDF weight scales each token's contribution by its relative importance in the document. The per-token attribution score averages the three lookups with a global baseline $\bar{f}$; scores above zero indicate association with above-average model predictions and scores below zero indicate the opposite.

1.2 Image Classification

Each image $I^{(n)}$ is a sample; each pixel location (i, j) is a feature. Pixel intensities are discretized into R equal-width ranges forming the bin vocabulary. The pairwise structure is *spatially local*: pixel (i, j) is paired with each of its 8-connected neighbors $(\Delta x, \Delta y) \in \{-1, 0, 1\}^2 \setminus \{(0, 0)\}$, yielding a 4-dimensional accumulator $\boldsymbol{W}[i, j, \mathrm{pos}, r]$ indexed by pixel position, neighbor direction, and neighbor intensity range. The outcome $Y^{(n)} \in \{0, 1\}$ is the binary image label. At inference, each pixel's attribution score is the mean of its 8 neighborhood conditional means retrieved from the accumulator, producing a spatial attribution mask.

1.3 Retrieval-Augmented Generation

A RAG pipeline maps a user query q through a sentence transformer ϕ to produce an embedding $\phi(q)$, then ranks a knowledge base of text chunks $\{d^{(n)}\}_{n=1}^{N}$ by cosine similarity $Y^{(n)} := \cos(\phi(q), \phi(d^{(n)}))$, returning the top-$k$ chunks as LLM context. To apply the Barn Effect, treat each chunk $d^{(n)}$ as a sample: its tokens occupy the feature columns and $Y^{(n)}$ is the similarity score. Training $\mathbf{\Phi}$ on the knowledge base produces token-level Barn attributions $\delta_c^{(n)}$ for every token c in every chunk n. A positive $\delta_c^{(n)}$ indicates that token c's co-occurrence context is associated with above-average similarity to the query; a negative value indicates below-average contribution.

1.4 Multi-Modal (Overview)

Treat modalities as disjoint column groups sharing a vocabulary; column pairs span within-modality and cross-modality combinations, and a single $\mathbf{\Phi}$ is learned over all pairs jointly. The gap-graph construction $\boldsymbol{G} = \mathbf{\Phi}_{\text{data}} - \mathbf{\Phi}_{\text{model}}$ applies: comparing interaction graphs trained on different signals can reveal cross-modal alignment gaps. A detailed treatment of the embedding case is given in Section 2.

2 Multi-Modal Embedding Extensions

The domain extensions above assume that each modality provides scalar or discrete feature values that can be binned directly. A more general construction applies when samples are described by dense embeddings from two or more modalities (text encoders, image encoders, structured attribute encoders, sensor embeddings), and the relevant co-occurrence structure lives in the *embedding space* rather than in raw values.

2.1 Encoding via Cluster Assignment

Let modality $m \in [M]$ provide an embedding $\boldsymbol{e}_m^{(n)} \in \mathbb{R}^{d_m}$ for each sample n. Train a clustering model (e.g., k-means) on the training embeddings for each modality independently, yielding K_m cluster centroids $\{\boldsymbol{\mu}_k^{(m)}\}_{k=1}^{K_m}$. Define the bin labeling function for modality m as the nearest-centroid assignment:

$$\ell^{(m)}(\boldsymbol{e}_m^{(n)}) \;:=\; \arg\min_{k \in [K_m]} \big\| \boldsymbol{e}_m^{(n)} - \boldsymbol{\mu}_k^{(m)} \big\|_2.$$

This reduces each modality to a single categorical column with K_m labels. The global vocabulary $\mathcal{V}$ is the union of all cluster-index sets; because labels are column-specific, cluster index k in modality a and cluster index k in modality b are distinct integers in $\mathcal{V}$. The two-phase algorithm then proceeds without modification: Phase 1 is the clustering step, and Phase 2 accumulates co-occurrence statistics over the encoded cluster labels.

2.2 Target Options

Two natural choices for the outcome $Y^{(n)}$ are:

- **Similarity target.** Set $Y^{(n)} := \cos(\boldsymbol{e}_a^{(n)}, \boldsymbol{e}_b^{(n)})$ for a designated pair of modalities (a, b). The interaction graph $\mathbf{\Phi}$ then encodes which cross-modal cluster combinations co-occur with high cosine similarity, revealing which embedding-space regions of each modality tend to be mutually aligned. $\bar{\delta}_m$ ranks each modality's cluster structure by its contribution to cross-modal alignment; $\delta_m^{(n)}$ identifies per-sample alignment anomalies.

- **Labeled target.** Set $Y^{(n)} \in \{0, 1\}$ or $Y^{(n)} \in \mathbb{R}$ to a ground-truth task label or score. The interaction graph then encodes which cross-modal cluster co-occurrences predict the label. Appropriate for multi-modal classification tasks (visual question answering, audio-visual sentiment, medical image + clinical notes) where a supervised signal is available.

2.3 Interpretation

An entry $\Phi_{u,v}$ where $u \in [K_a]$ is a cluster of modality a and $v \in [K_b]$ is a cluster of modality b gives the log-frequency-weighted conditional mean of $Y^{(n)}$ over all samples whose modality-a embedding falls in cluster u and modality-b embedding falls in cluster v. The attribution $\delta_m^{(n)}$ measures how much that sample's cluster context deviates from the global mean target, aggregated over all cross-modal and within-modal pairs involving modality m.

Remark 1 (Cluster granularity tradeoff)**.** *Increasing K_m produces finer-grained bin labels at the cost of sparser $\mathbf{\Phi}$ (fewer samples per cluster pair). Decreasing K_m yields denser accumulation but coarser embedding-space regions. The significance threshold naturally suppresses cluster pairs that co-occur too rarely for reliable estimation, providing an automatic quality floor without requiring K_m to be tuned independently.*

Remark 2 (Cross-modal gap analysis)**.** *Train $\mathbf{\Phi}_{\text{sim}}$ with a similarity target and $\mathbf{\Phi}_{\text{label}}$ with a ground-truth label on the same dataset and cluster schema. The element-wise difference $\boldsymbol{G} = \mathbf{\Phi}_{\text{sim}} - \mathbf{\Phi}_{\text{label}}$ identifies cross-modal cluster combinations where embedding alignment and task-relevant signal diverge: large positive G_{uv} means clusters align well in embedding space but do not predict the label; large negative G_{uv} means label predictability without geometric alignment. This serves as a targeted diagnostic for multi-modal retrieval and classification pipelines.*